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Highlighting this over that – a cognitive and frequency explanation of transitive variation

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ABSTRACT

The crosslinguistic variation of the transitive clause has been explained in different ways. This study approaches it from a mathematical perspective employing a scorecard to predict the high frequency of the SOV and the SVO over other possible types. A matching theoretical framework is found in markedness. It is proposed that the attested typological distribution pattern depends on the similarity of subject (S) and object (O)—both being nominals—and on their inequality because the subject, appearing in intransitive and transitive clauses alike, is more frequent in language use than the object, which is absent in intransitive clauses. Thus, the simplest explanation of the distribution is that such statistical asymmetry gives rise to the preference of highlighting the secondariness of the object in the basic order. SO is generally preferred over OS. Additionally, early SO is preferred over late SO, and late OS is preferred over early OS. The whole six-way distribution of subject, object, and verb (SOV > SVO > VSO > VOS > OVS > OSV) is generated on this basis.

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1. Introduction

The dominant word order of the transitive clause is perhaps the single most frequently cited feature in typological literature. However, it is still to be established why some orderings of SUBJECT (S), OBJECT (O), and VERB (V) are utterly common and others strikingly rare as a dominant order cross-linguistically. It has been suggested that statistical biases may depend on biological and psychological universals, on all or most languages having a common ancestor, or on how languages evolve and spread.

When a new language arises from a parent language, the most reliable predictor is inheritance: in 78 percent of the cases, the new language will

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have the same order as its parent language (Hammarström 2015, 43).¹ Examples are found in the Anglo-Celtic Isles where Scots, deriving from Middle English, has the same SVO order as English. Celtic languages, including Cornish, Irish, Manx, Scottish Gaelic, and Welsh, derive from a common proto-language with VSO and maintain its order. Such is not always the case. Modern Romance languages including French, Italian, and Spanish have SVO despite being derived from Latin, which has a free word order with a high frequency of SOV (Bauer 2009, 268).

The effect of language contact is estimated at eight percent (Hammarström 2015, 43). Many Uralic languages, for example, Estonian and Finnish, have prehistorically shifted from SOV to SVO, which is the word order of their neighbouring languages, including Latvian, Russian, and Swedish. By contrast, Basque (SOV, isolate) has not adopted SVO order from its Romance neighbours.

There appears to be an unidentified factor accounting for the remaining 14 percent of dominance in the transitive clause. Hammarström (2015, 27–32, 43) concludes that a universal force modifies the long-term outcomes of genealogical and contact effects, leading to the observed distribution pattern (Figure 2). Hammarström bases his conclusion on his large data analysis, representing the classical UNIVERSALIST view (3.1).

What qualifies as a theory to explain such an unidentified factor? This paper, representing research in theoretical linguistics, does not aim to present new

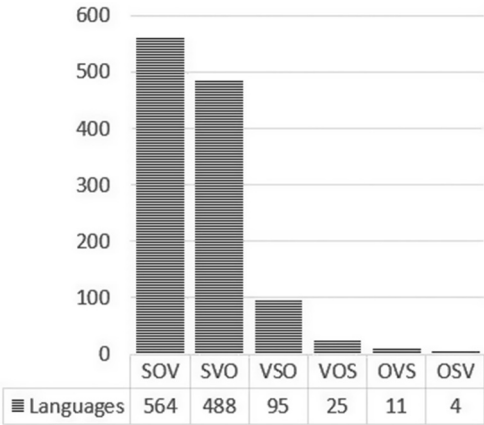


Figure 1. The crosslinguistic transitive distribution. A dataset (Dryer 2013b) demonstrates the transitive distribution based on the number of languages in a given sample (NO-DOM excluded).

¹Hammarström’s presentation slides (Hammarström 2015, < https://web.archive.org/web/20220811222639/https://www.eva.mpg.de/fileadmin/content_files/linguistics/conferences/2015-diversity-linguistics/Hammarstroem_slides.pdf>) offer a more comprehensive analysis of his database research than his published paper (Hammarström 2016).

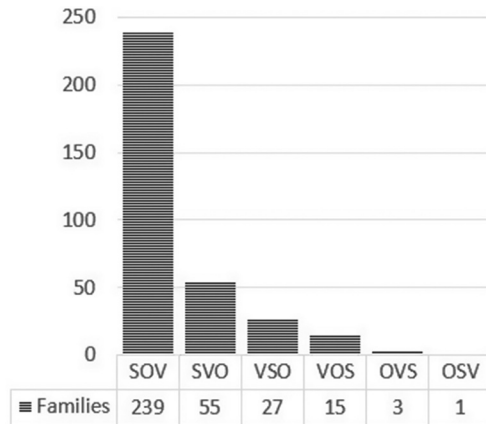


Figure 2. A dataset (Hammarström 2016) demonstrates that when language families and isolates are counted, SOV is by far the most frequent type globally. As in Figure 1, NO-DOM languages are excluded.

empirical research—the necessary material already having been made available—but offers a new model of explanation (Section 4). This model is developed in the spirit of the pragmatic and humanistic² tradition of the Prague linguistic circle and its derivatives, including Systemic Functional Linguistics (Halliday and Matthiessen 2014), Functional Discourse Grammar (Hengeveld and MacKenzie 2008), and others such as SILF (*Société Internationale de Linguistique Fonctionnelle*) and Danish Functional Linguistics (Harder 2005).

Hjelmslev ([1943] 1969, 1963) envisioned linguistic typology as the search for the essence of human language via an exhaustive structural comparison of the world’s languages. In his concept, typology and general linguistics are part and parcel of the same endeavour:

It is not the individual language alone that is the object of the linguist, but the whole class of languages, the members of which are connected with each other and explain and cast light on each other. It is impossible to draw a boundary between the study of the individual linguistic type and the general typology of languages; the individual linguistic type is a special case within that typology and, like all functions, has its existence only by virtue of the function that connects it with others. (Hjelmslev [1943] 1969, 126)

In the following sections, the data and the problem are presented (2). Previous attempts are introduced and critically assessed (3), followed by the introduction of a functional solution (4). Concerns regarding the utility

²The Prague concept of structural and functional linguistics is influenced by Ferdinand de Saussure’s sociological perspective of linguistics and by Edmund Husserl’s phenomenology. A standpoint shared by the Prague and the Copenhagen school is in their instrumental view of language, which is described as “humanistic” by Daneš (1987, 29–30fn).

of transitivity-based typology are addressed (Table 1 and Section 4.3). Empirical research related to the proposed solution is discussed (4.4).

2. Universals, the data, and the problem

A classic example of a transitive expression in English is the SVO sentence “*John loves Mary*”. It is a predicate-argument structure with two arguments, as in $P(x,y)$. The three members are traditionally called subject, verb, and object, although it would be more correct to call them agent, event, and patient because typological generalizations are made on a semantic basis. As Dryer (2013b) comments on his criteria:

The terms subject and object are used here in a rather informal semantic sense, to denote the more agent-like and more patient-like elements respectively. Their use here can be defined in terms of the notions S, A, and P, where the S is the single argument in an intransitive clause, the A is the more agent-like argument in a transitive clause, and the P is the more patient-like argument in a transitive clause. [...] The term subject is used for the A while the term object is used for the P. A language [with] SOV could thus also be equally well and perhaps more accurately described as APV. Note that many linguists use the terms subject and object somewhat differently from this, and some linguists question the applicability of these terms to some languages, but these issues do not arise with the use of these terms here. (Dryer 2013b)

Typological examples of the six dominant transitive orders include the following.

(1) Hindi (SOV)

उमा	कमीज़	लाई।
<i>uma:</i>	<i>kami:z</i>	<i>la:i:</i>
Uma-NOM ³	shirt	brought

‘Uma brought a shirt.’ (Koul 2008, 38)

(2) Mandarin (SVO)

爸爸	爱	妈妈
<i>bàba</i>	<i>ài</i>	<i>māmā</i>
dad	love	mom

‘Dad loves mom.’ (personal knowledge)

(3) Standard Arabic (VSO)

الدرس	الطلاب	كتب
<i>kataba</i>	<i>t-tulaab-u</i>	<i>d-dars-a</i>
wrote.3MSG	the-students-NOM	the-lesson-ACC

‘The students wrote the lesson.’ (Jouini 2018, 234)

(4) Malagasy (VOS)

misotro ronono izahay
 drink milk we
 ‘We are drinking milk.’ (Keenan and Ochs 1979, 119)

(5) Panare (OVS)

arakon pétyuma-ya’ apo’
 black.monkey hit-PST man
 ‘The man hit the monkey.’ (Gildea 1989, 13)

(6) Nadëb (OSV)

awad kalapéé hapúh
 jaguar child see.IND
 ‘The child sees the jaguar.’ (Weir 1994, 309)

Any language may make use of a variety of orderings but is considered to have a dominant order if an ordering is attested in texts at least twice as frequently as any other (Dryer 2013a). On such a basis, a number of languages are left without a dominant order. Some of them are considered to have two dominant orders based on various criteria as discussed in detail by Dryer (2013b; see also 4.3). Well-known examples from Europe include Dutch, German, and Hungarian (SOV with SVO), and modern Greek (SVO with VSO). Based on Dryer (2013b), Belarusian has no dominant order⁴, and many languages around the world are characterized as being discourse-configurational rather than having a fixed word order (Mithun 1992, 58–59; K. Kiss 1995).

⁴My information on Hungarian and Belarusian is based on the maps relating to Dryer (2013b): 81A and 81B.

Nonetheless, languages lacking a single dominant order (“NO-DOM”) make up far less than ten percent of the data⁵ and are often put aside when looking for an explanation of the biases. The question of NO-DOM languages should be returned to after the main pattern has found its explanation, but it is outside the scope of the present paper.

From a purely logical point, any order should be just as good as the other. Predicate logic uses VSO order out of notational convenience, but it is not purported to represent any logical reality. Furthermore, even though there are free word-order languages, and many languages attach affixes replacing personal pronouns to the verb, each attested full transitive clause must have materialized with one of the six logically available orderings: SOV, SVO, VSO, VOS, OVS, or OSV, with V representing the finite verb. Since there is no reason from a logical point why any of the orderings should be superior or inferior, their distribution is expected to be random by default, each landing at approximately 16.7 percent or 1/6. The actual statistics are far removed from that, necessitating an explanation of the biases (cf. Comrie 1981, 201). Issues concerning the criteria for the data analyses in question have been discussed by, e.g., Stassen (2010) and Frey (2015, 515–518).

2.1 *Adjusting for a Colonial effect*

Since a majority of linguists speak English and other European languages with SVO, it may strike as odd that SOV should be the most common order cross-linguistically. It is nonetheless well established. The distribution hierarchy, from the most to the least frequent, is SOV > SVO > VSO > VOS > OVS > OSV. The frequencies of SOV and SVO are relatively close. Hammarström (2015, 33) explains it to be an effect of colonialism. Hundreds of languages have vanished in the Americas—especially on the Brazilian east coast—and Australia. The effect is much weaker in Africa and Asia, which form a landmass with Europe.

It is well established that diseases contributed to the downfall of hundreds of native nations in the new continents (Cook 1998). The effect of colonialism has been far more moderate in Africa and Asia, where higher genetic resistance to pandemics had already evolved, aiding the survival of nations and their languages. For example, it is suggested that the bubonic plague, before decreasing populations in Europe, originated in Asia and spread via trade routes (Morelli et al. 2010). In direct or indirect contact with these routes, there are today different types of word-order hotspots that withstood the subsequential colonial period, namely, SOV hotspots in New

⁵According to Hammarström (2016, 25), 2.3 percent of individual languages or 6.1 percent of language families are known to be NODOM. The actual share of each is expected to be about one point higher because the data of the transitive order was missing from 30 percent of the full set of individual languages versus 14 percent of the language families.

Guinea and the Indian subcontinent and SVO hotspots in Sub-Saharan Africa and mainland Southeast Asia (Dryer 2013b).

At the same time, there are conspicuous empty spots in the Caribbean, continental Americas, Australia, Tasmania, and New Zealand. Based on areal linguistics and what is known from historical sources, these areas previously included non-SVO hotspots. According to Hammarström's (2015, 11–12, 33, 2016, 20–21) analysis, it is their disappearance that explains the relatively high frequency of SVO in comparison with all other types. To be clear, colonialism did not significantly increase the number of SVO languages as the spread of English, Spanish, Portuguese, and others does not per se add to the total number of SVO languages. Rather, it is a historical accident that they would mostly wipe out the other word-order types, especially SOV as seen by a comparison of Figures 1 and 2.

For this reason, Hammarström (2015, 2016; cf. Figure 2) conducts a stratified analysis to uncover a precolonial distribution, which turns out to be heavily dominated by SOV, failing to corroborate a similar tendency for SVO. In the present paper, the transitive data or distribution refers to Figure 2 unless otherwise stated.

3. Previous research

Greenberg (1963a) presented a set of transitive data as part of his groundbreaking research on linguistic universals. He based his cross-linguistic analyses on markedness theory as advocated by Roman Jakobson (Battistella 2015). Jakobson's (1963, 268–269) explanation of the dominant word order of the transitive clause is that the object, when fronted, is perceived as emphatic. Thus, SO is preferred over OS as the basic, unmarked order cross-linguistically.

Jakobson exemplified his universal claim by discussing Russian word-order variation. The transitive expression “Lenin cites Marx” can appear in all six possible orders with varying degrees of morphological markedness. The SVO sentence *Lenin citiruet Marks***a** marks the object with the –a ending and can be transformed to any of the other five orderings. Additionally, having the same order as the English passive sentence “Marx is cited by Lenin” (SVO but patient-verb-agent), Russian also has the dually morphologically marked sentence *Marks citiruetsja* **Leninym**, which can likewise be transformed into any of the other five orderings. Jakobson observes that in each case, the basic order places the morphologically unmarked participant (*Lenin* or *Marks*) at the beginning of the sentence.

Making his remarks at the seminal 1961 Dobbs Ferry conference, which marked the beginning of a new era in typology, Jakobson (1963, 269–270) concluded that “The nearly universal precedence of the subject with regard to

the object, at least in unmarked constructions, points to a hierarchy of focusing.” He argues that the zero affix is universally unable to be steadily assigned to the marked category and, conversely, that the real affix cannot be steadily assigned to the unmarked category. To elucidate this, Jakobson linked conceptual markedness (‘categories’) and overt markedness (‘affixes’), but did not intend it as an absolute universal:

language **tends to avoid** any chiasmus between pairs of unmarked/marked categories on the one hand, and pairs of zero/non-zero affixes (or of simple/compound grammatical forms) on the other hand. (Jakobson 1963, 270; emphasis added)

Jakobson (1963, 270) also suggested that there is an unmarked focus at the beginning and a marked focus at the end of the sentence. His idea was related to the functional sentence perspective (FSP) of the Prague linguistic circle, which borrows from Henri Weil’s (1844) crosslinguistic research on word order. According to the FSP—in terms of topic versus comment, and given versus new—COMMUNICATIVE DYNAMISM gradually rises from the beginning to the end of the sentence. Thus, in the SVO order, analyzed in terms of INFORMATION STRUCTURE and INFORMATION STATUS, the subject tends to be the least dynamic and the verb intermediate, while a contextually independent object has the most communicative dynamism (Firbas 1971, 137; 1987; Davidse, Van Praet, and Njende 2019). To accommodate my solution in (4), Firbas’s concept might permit the rewording that the subject is unfocused, the object is focused, and the verb is neutral.

Greenberg found markedness theory instrumental for uncovering linguistic universals and built his implicational hierarchies on the concept. As Newmeyer (2001) observes,

The influence of the PS [Prague School] permeates every page of Greenberg’s seminal paper. Indeed, by Greenberg’s own acknowledgement (Greenberg 1963[. . .]: 104), the paper was written in response to Roman Jakobson’s call for an ‘implicational typology’ of language universals [. . .]. PS terminology is also rampant in the Greenberg paper, as is evidenced by the frequent description of one order of elements as being ‘more marked’ or ‘less marked’ than another. (Newmeyer 2001, 111)

Jakobson’s formulation of implicational hierarchy was as follows: “[T]he presence of *A* implies the presence (or on the contrary the absence) of *B*.” (Jakobson 1962, 526). To explain the emergence of such hierarchies, Jakobson proposed universal laws steering language change, based on his Saussurean-inspired theory of distinctive opponents. However, Jakobson emphasized that such laws can only be fully understood after extensive empirical investigations, which required lifting a taboo on typology that had been posed by nineteenth-century grammarians, such

as Schlegel, who exaggerated the uniqueness of all languages (Jakobson 1962).

Zipf's LAW OF ABBREVIATION (Zipf 1936) was also known to linguists at the time. Referring to it, Trubetzkoy (1939, 262) suggested frequency as a factor determining markedness. Osgood (1963) proposed the following generalization at Dobbs Ferry:

At all levels of units in a language, the competing alternatives will be organized hierarchically in terms of frequency of occurrence and with a relatively low-entropy distribution approximating the Zipf function. (Osgood 1963, 240)

Thus, the subject, being the more frequent item, would be statistically shorter than the object. This prediction holds. The underlying explanation is that it simplifies language making it more ECONOMIC to produce and process (cf. Vicentini 2003). Still, markedness left the explanation of the basic transitive order incomplete. Jakobson's pragmatic-communicative principle correctly predicts the well-known SO > OS preference but has not been extended to an account of the full tripartite pattern until this paper (Section 4).

3.1 *Classical universalist explanations*

Inspired by Greenberg's work on linguistic universals, different predictive algorithms and explanatory models of the transitive distribution have been proposed since the 1970s. PREDICTION is used with a broad meaning in this context because the purpose is to find variables that generate the distribution, which is now known to be SOV > SVO > VSO > VOS > OVS > OSV, indicating the order from the commonest to the least common. However, the existence of OVS and OSV languages was initially unknown and subsequently controversial for a long time (e.g., Polinskaja 1989). Thus, the purpose of the 1970s models was to explain why some orderings may be alien to the human mind.

Greenberg's sample included only three types: SOV, SVO, and VSO languages although well-known examples of VOS include Malagasy, the main language of Madagascar with 25 million speakers, which was added by Ultan (1969). Several generativists (see Bach 1974; Culicover and Wexler 1974; Schwartz 1974; Diehl 1975; Pullum 1977; Emonds 1980) proposed sets of transformational rules to account for the absence of the remaining logically possible orderings. However, Derbyshire (1977), and Derbyshire and Pullum (1981) reported the discovery of OVS and OSV, making previous proposals obsolete.

New algorithms were proposed, but there is a fundamental discrepancy between the expectations and the empirical data. In classical grammar, the sentence is divided into subject and predicate, conflating the TOPIC with the subject and the COMMENT with the predicate (Haberland and Nedergaard

Thomsen 1994, 181). In the Ancient and Mediaeval periods, the purpose was to show how language is a rational tool for making propositions about the subject (*‘Iron is hard’, ‘Cows eat grass’, etc.*; cf. Seuren 1998).

The reason why generativists and typologists still consider the subject-predicate divide elementary is unknown. Probably the most transparent and authoritative discussion is provided by Chomsky (1965, 73–74, 116–117), who wonders why the sentence has to be reconstructed in such a way and proposes Generative Grammar as a research program that might one day find out. Decades later, as the ERP (Event Related Potential) imaging technique became available, Chomsky prompted neuroscientists to examine how structural assumptions made by Generative Grammar thus far were manifested in the brain. Kluender and Kutas (1993) found no sign of grammar working along such fixed pathways (cf. Figure 3) and concluded that sentence processing is, rather, a matter of the interaction of syntax and semantics. The issue continues to arise, e.g., with Sprouse and Lau (2013), who cite Kluender and Kutas (1993), implying that it is evidence for Generative Grammar. Sprouse (2023), rather than being troubled by the lack of insight, proposes that figuring out the link between syntactic theory and the brain “will be up to the next generations of syntacticians” (p. 24). This neatly summarizes the information extractable from the literature.

Greenberg (1963), for his part, stated: “It is here assumed, among other things, that all languages have subject-predicate constructions” (p. 59). Such a statement could be interpreted in different ways, but BRANCHING, a fundamental concept in generative and typological syntax, builds on the subject-predicate divide (cf. T. Kiss and Alexiadou 2015). In this context, Hawkins (2004, 174) likewise considers Kluender and Kutas (1993) as evidence for his model based on the subject-predicate split.

As discussed by Song (2012), Dryer’s (1992) BRANCHING DIRECTION THEORY draws evidence from OV/VO and SV/VS typology. However, as we already saw in Figure 2, neither of these variables appears to bear on the transitive clause. SV is the most and least common type and, as such, might not have explanatory potential. Regarding OV/VO, the most common type (SOV) is OV and the second-commonest (SVO) is VO, ruling each other out. Thus, what remains with explanatory potential is just $SO > OS$, but it is not relevant to branching theory. Nevertheless, Dryer (1997) explicitly calls for combining SV/VS and OV/VO to circumvent the six-way typology.

Let us examine how the combination of SV/VS and OV/VO predicts the transitive distribution based on permutational distances. Traditionally, SVO is hypothesized to be the initial state, and the whole distribution is predicted based on the closeness of other orderings to it. Let us start by making the tiniest possible word-order change or MOVE, which is flipping the VP to arrive at $SVO > SOV$, which is an incorrect prediction. Figure 3 corrects the algorithm by

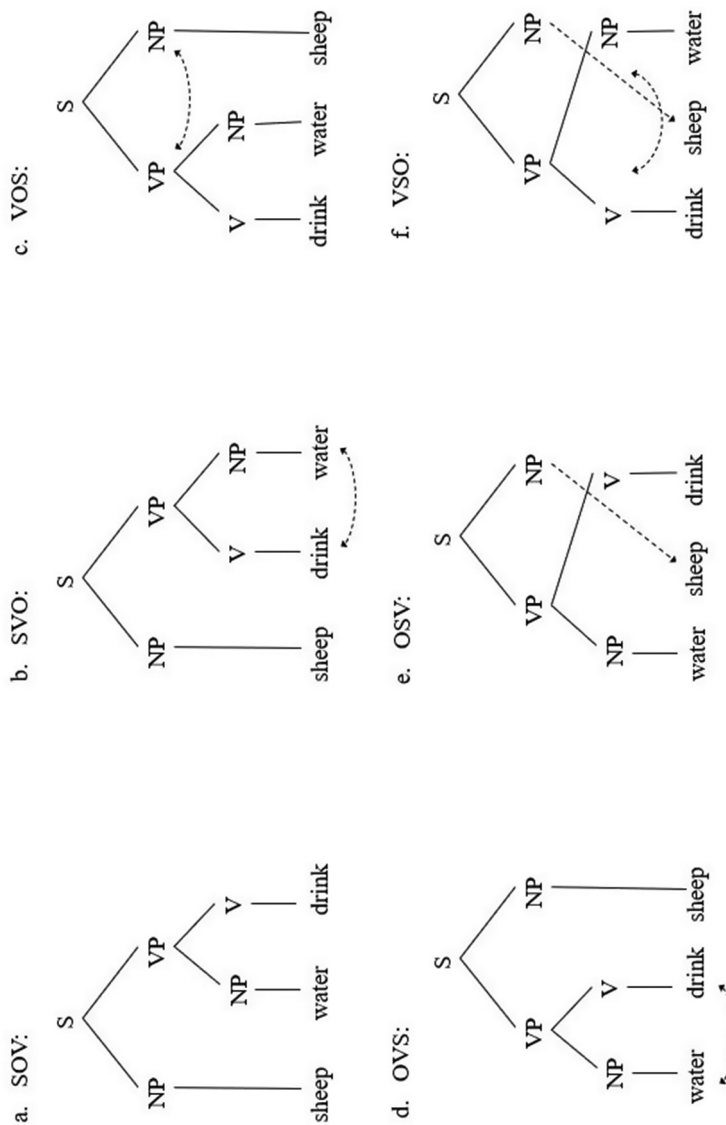


Figure 3. Deriving a prediction based on permutational distances from SOV. Arrows indicate move operations. As seen in the figure, Generative Grammar, and branching typology fail to predict the transitive distribution correctly, both presuming an underlying universal subject-predicate (NP VP) structure. In the given context, the subject-predicate split relates to SV/VS typology, while the verb-object (V NP) split within the predicate (VP) relates to OV/VO typology. If the subject-predicate split was a cognitive and typological reality, OSV and VSO might not exist as suggested by (e) and (f), where the purported ‘predicate’ (VP) is broken.

starting from SOV instead, yielding SOV (a.) > SVO (b.), as is correct. The second move or permutation is a predicate-subject flip (c.), which remains undisruptive regarding the NP-VP divide. It predicts that VOS (c.) is the third-commonest ordering, which is incorrect. It should be VSO (f.).

In sum, an explanation based on the classical subject-predicate split fails. This realization would either call for the rejection of Generative Grammar and mainstream syntactic typology, or the data. Consequently, there are two dominant paradigms about syntactic universals, which share a disproven hypothesis. Generative Grammar coped with the lack of empirical support by declaring that syntactic universals are irrelevant to a hypothesized innate universal grammar (Newmeyer 2004), which might sound self-contradictory. Typologists, led by Dryer, coped similarly by declaring that the six-way distribution is irrelevant to word-order typology.

Both cases exemplify *SELECTIVE REPORTING*, by which I mean that there is indeed a vast research literature on branching and processing but it fails to discuss the very basics of the issue. What is reported is the success of syntacticians in providing research on universals, while discussion of its failures is omitted, like the problem in Figure 3, which is obvious information but not made explicit by previous papers. Although ignored, this is not the first time selective reporting in linguistics has been taken up. Schwarzfriesel (2012) asks why scholars representing different camps of ‘cognitive linguistics’ filter out information that challenges their preconceptions, losing sight of the big picture.

Dryer (1997) enumerates eight reasons in support of his stipulation to discard the six-way distribution and keep SV/VS and OV/VO. In Table 1, each item and its objection are presented.

As is seen from Table 1, Dryer’s reasons for abandoning OS/SO are not convincing. Put in simple terms, SO/OS and the six-way distribution are significant, but Dryer proposes replacing them with SV/VS and OV/VO, which, from the topic’s perspective, are insignificant. To wit, such an arbitrary stipulation is what keeps the models proposed by generativists and typologists from collapsing. Dryer (2013c) maintains that the six-way typology must be abandoned but admits that SV/VS has failed to provide insight.

To continue with proposals that nonetheless cater to the six-way distribution, it remains a common problem with the published algorithms and models that they make the wrong predictions. An alternative way to address the problem was made via semantics (see Maxwell 1984), which suggests VSO is the most favoured type. The 1980s also saw proposals that combine three pragmatic principles into a unified explanation, e.g., Mallinson and Blake (1981) and Krupa (1982). The only influential attempt from this period was Tomlin (1986), who improved on the earlier three-way models in conjunction with his proposal for a new six-way hierarchy: SOV = SVO >

Table 1. Arguments for abandoning transitivity as a type indicator. Dryer’s (1997, 69–70) eight arguments (middle column) and their rejection (right column) are presented.

#	Against S/O/V typology (Dryer)	Objection (Austin)
1	“[SV/VS and OV/VO] allows easy classification of languages which are indeterminately VSO/VOS.”	Easiness is not a counter-argument against S/O/V as a whole, and recent research continues to support a distinction between VSO and VOS (e.g. Hammarström 2015; 2016). Conflating VSO and VOS might appear to help the subject-predicate split, but as demonstrated by Figure 3, it cannot fix it—unless VSO is removed from consideration.
2	“there is no evidence that the difference between VSO and VOS languages is predictive of anything: the properties that are typical of VSO languages are apparently also typical of VOS languages, and hence VSO and VOS are best treated as belonging to the same type.”	As an example of a behavioural difference, topicalization by SO→OS inversion is only predicted for VSO as discussed in (4.3).
3	“the difference between VSO and VOS order is a relatively unstable one, both orders being commonly found as basic orders within the same language family and within the same linguistic area.”	Basic VOS is not found in Celtic and Semitic languages. Generally, though, linguistic areas often exhibit basic-order variation.
4	“the traditional six-way typology is based on a clause type that occurs relatively infrequently, while the proposed typology is based on clause types that occur much more frequently.”	What might such a ‘more frequent’ type be if not the transitive clause? OV/VO, as proposed by Dryer, relates to it, too. As seen in Figure 3, the two pairs (SV/VS and OV/VO) generate the six-way distribution albeit erroneously.
5	“there are many languages which have word order sufficiently flexible that they are impossible to classify by the traditional typology but which are still classifiable by the proposed typology”	This is a common situation in typology. If Dryer’s #5 is accepted, many more established data have to be rejected. For example, many languages employ a dual strategy for relativization.
6	“there are other languages with word order even more flexible but which are still classifiable either for the order of subject and verb or for the order of object and verb”	The unlikelihood of dominance in the three-partite account as opposed to pairs is simply mathematically predictable, and yet, statistically significant differences are established against such an elevated improbability. That only demonstrates even more strongly that they are no coincidences.
7	“the proposed typology is superior because it isolates the order of the object and verb, the more fundamental typological parameter in terms of word order correlations”	SV/VS does not isolate OV/VO: the three, including SO/OS, are all variables, and we are not told why OV/VO should be combined with SV/VS rather than SO/OS. Thus, rejecting SO/OS might reject SV/VS all the same, justifying the overall rejection of branching theory either way. In consequence, Dryer’s #6 is not a wise argument.
8	“[The six-way typology] overlooks the order of subject and verb in intransitive clauses, even though the order in such clauses is occasionally different from the order of subject and verb intransitive clauses, and even though intransitive clauses containing a noun subject are much more common than transitive clauses containing a noun subject”	“Occasionally” does not cater to a sufficient counter-argument and relates to SV/VS all the same. Dryer suggests that the subject, rather than the object, is often pronominal in transitive clauses, whereby it is often suffixed in the verb, but such variability is not limited to the transitive. Essentially, all languages must include a way of expressing non-pronominal third-person transitives, e.g., ‘The fox chased the rabbit.’ Here, the relevant question is how such clauses are ordered cross-linguistically, and the task of S/O/V typology is to answer that question rather than some other question.

VSO > VOS = OVS > OSV, with the equations indicating equal frequency. The equations were rejected by Dryer (1989, 260, 270; Song 2012, 27), based on a bias in Tomlin's sample.

Nevertheless, Hawkins's 'efficiency' approach is based on Tomlin's distribution. Hawkins (1994, 2004) incorporates the problem into his broader framework of EARLY IMMEDIATE CONSTITUENTS, a version of branching theory, linking the proposal with issues of phrasal length. According to it, VO languages (like English) exhibit increasing length throughout the sentence, while OV languages (like Hindi) do the opposite. However, Song likewise calls this solution into question because Hawkins claimed that SOV languages have longer subjects than objects, but Song (2012, 247) suggests that the provided evidence is not representative of the whole data.

Manning and Parker (1989) made the exceptional correct prediction SOV > SVO > VSO > VOS > OVS > OSV with a noticeably complex proposal (see 4.3), and their eclectic assortment of causal explanations attracts a dedicated critique from Peeters (1991). The model has fallen into oblivion even though, technically, it is still better than the rest. More recent attempts, including Cysouw (2008) and Maurits, Perfors, and Navarro (2010), employ computational techniques to approach the problem from a universalist perspective but without the right predictions.

3.2 *Evolutionary explanations*

The problem of syntactic universals has also been approached from a historical angle. It might be that cross-linguistic patterns reflect the way languages have evolved. According to Croft (2003),

the integration of historical relations between languages in inferring language universals requires a shift to analyzing language structure as part of a dynamic evolving linguistic system over time, and language universals as universal constraints on the evolution of linguistic systems. This shift in approach, necessitated by methodological issues in constructing language samples, actually dovetails with the general shift in typological theory from universals of synchronic language systems to universals constraining language change[.] (Croft 2003, 6)

However, as was pointed out by Jakobson (1962), the problem with universalist typology before the Dobbs Ferry conference was that it had been banned by nineteenth-century grammarians, who precisely supplanted it with an evolutionary approach. Thus, it is undeniable that linguistics is going around in circles.

On the positive side, computational techniques were adopted in the early twentieth century to analyse very large sets of data. It is, however, noteworthy that Hammarström's published article (2016) presents such a project in a positive light, but his slides discuss several problems in the way the data

have been interpreted by various scholars. In his conclusion, Hammarström (2015) makes a wholesale dismissal of the evolutionary approach to transitive typology.

To be fair to both parties, Croft (2003, 17) refers to a link between grammaticalization patterns (diachrony) and word-order patterns (synchrony) and mentions the case of adpositions. In brief, there is an established link between OV/VO and the order of the noun and the adposition (i.e., prepositions versus postpositions) but the question remains whether a shift in the first propels a shift in the second or vice versa. Still, as research establishing a causal link has not been made available, it only leaves us with Hammarström's criticism, especially since none of the evolutionary attempts has produced the correct predictions.

How the evolutionary approach is different from the universalist approach is not completely clear. Like Givón (2018) and Bickerton (1981), Christiansen and Devlin (1997), too, support *a priori* learning and processing conditions as causes of the hypothesized initial stage. They subscribe to Hawkins (1994) which, as we learned in (3.1) is a universalist approach even though it is frequently cited in the evolutionary frame of reference. The fact also remains that Tomlin's distribution, which Hawkins is based on, was rejected. His model, like Dryer's, is an approach to branching theory, which assumes the subject-predicate divide. Even though both draw from typology, the dichotomy is not supported by the transitive data (Figure 3). In sum, it is not clear that there has been any progress since the return to evolutionary linguistics or that there will be any in the future.

An underlying idea of an SOV Proto-World language may have some intuitive value. Gell-Mann and Ruhlen (2011) argue that languages turned SVO more lately, and there is now a shift to VSO and VOS in process. Indeed, Indo-Aryan languages, including Sanskrit, have SOV, as do most Iranian languages. Following an Indo-European expansion in Western Europe, Latin had SOV, but the subsequent vernaculars (Italian, French, Spanish, etc.) have SVO and, at the west end of the European map (Dryer 2013b), there are Celtic languages with VSO. But Hammarström's (2015) conclusion is different. He points out that there were a larger number of Celtic VSO languages until they were replaced by SVO languages. Thus, there is no indication of Europe turning VSO.

Moving eastward from SOV-dominant India, we find a mainland Southeast Asian hotspot of SVO languages, followed by a human expansion into the Pacific with many V-initial languages including Rapanui (Easter Islands, VSO) at the east end of the Austronesian language map (Dryer 2013b). By contrast, Australian data does not support the notion of SOV evolving into SVO and then VSO following migration from north to south.

Historical evidence for an SOV Proto-World is controversial and has, for instance, been resisted by Chinese historical linguists, who find no support

for the claim that Chinese had SOV before shifting to SVO (Chappell, Ming, and Peyraube 2007). The estimated time span is also not favourable. Based on archaeological evidence, human language may have emerged as long as 50,000 years ago (Gell-Mann and Ruhlen 2011), and some estimate it to be up to 350,000 years old (Perreault and Mathew 2012). The oldest surviving pieces of some form of writing, albeit without transitive clauses, are less than six thousand years old (Chrisomalis 2009). This means that the amplitude of language change lending itself for observation and reliable reconstruction could have occurred manifold in prehistory. There is no guarantee that any relics from a Proto-Human language survive in the material, or that we could identify them as being such. Thus, it may all be a vanity project, a realization that was the starting point of Saussure (1916) over a hundred years ago.

Even assuming the Proto-World hypothesis, the explanatory value of the enterprise is called into question. Gell-Mann and Ruhlen (2011) appear to be proposing a kind of historicism: Rapanui has the order it has because it first had SOV; then it changed to SVO, and then it changed to VSO. In the absence of preserved data, we do not know whether this pattern is historically correct. If we assume it is, for the sake of argument, even then the claim does not seem to fully address the question “Why does Rapanui have VSO rather than some other order?”

For such reasons, Gell-Mann and Ruhlen (2011) are criticized by Givón (2018, 207fn), who based his original idea on language learning considerations. He speculates that childhood language learning patterns and primitive communication needs favour SV and OV, concluding that proto-human had SOV. However, it should not be ignored that the proposed pair likewise generates OSV unless one assumes the classical predicate-subject divide, which brings us back to the problem discussed in (3.1).

By contrast, Bickerton (1981) argues that Proto-World must have had SVO, which is better for disambiguation in the absence of highly evolved grammar including subject or object markers. Maurits et al. (2010), likewise, argue for the superiority of V-medial ordering (3.2), as does Ferrer-i-Cancho (2008). Ferrer-i-Cancho (2014) reviews several twenty-first-century historical and cognitive studies on the topic. His conclusion is that languages first had SOV, then turned SVO, and are now turning back to SOV but admits that experimental findings do not shed sufficient light on the matter.

Ferrer-i-Cancho (2015) proposes that, despite the functional value of V-medial, the low frequency of OVS is due to its long permutational distance from the original SOV order, which, again, brings us to the problem in Figure 3. But what if Proto-World was OVS nonetheless? For what it is worth, Maurits and Griffiths’s (2014) computer analysis points to an O-initial Proto-World, but they reconstruct it as SOV, guided by “expert opinion” (*sic*, p. 13577). It is an example of how presumptions guide results. Another example is Gell-Mann and Ruhlen (2011), who demonstrated that

languages never turn back to SOV via the elimination of any such data. Hammarström describes their analysis as “rigged” (2015, 27) and finds the real change pattern to be the same as the synchronic pattern (i.e., changes to SOV are the most common, changes to SVO the second-most, etc.).

Whatever the preferences and non-preferences, Trudgill (2011, 100–101) suggests that non-preferred types only thrive in small speaker communities because it is difficult to find wide acceptance for non-optimal orders. Trudgill borrows from Nettle (1999, 139), who argues that the rare O-initial languages have a small median speaker number.

However, S-initial languages, being the commonest ones, are by default expected to constitute the majority of all languages, large and small alike. For example, most of the endangered and moribund languages of Europe have SVO because it is typical in the area. Hammarström (2015) tested Nettle’s claim and found that when languages are sampled randomly, the median number of speakers of O-initial languages is actually slightly higher than that of SVO languages.

Finally, as already mentioned, Hammarström (2015) finds the effect of language contact, in terms of adopting the order of a neighbouring language, to explain eight percent of the observed dominant order variation. If it was much higher, obviously, all languages would end up with the same dominant order in the long run, but that is not suggested by the synchronic and diachronic data.

4. The solution of the transitive distribution

4.1 *Highlighting: a new typological generalization*

The generalization is made that **languages most frequently have early SO and most infrequently early OS**. SOV exemplifies early SO, and VSO exemplifies late SO, while SVO is intermediate between the two, based on Table 2. This notion is expanded into a prediction of the whole hierarchy when proposing $SO > OS$ and that the front-back asymmetry can be expressed in terms of decreasing numeric value by giving S or O two points in the initial position, one point in the middle position, and zero points in the final position. If the order is OS, the two constituents receive negative points. The reason for counting like this is that these operations generate the following scorecard, which generates the correct full six-way distribution (Table 2).

The novel prediction (Table 2) is generated by the following four rules:

- (i) Each position in the transitive clause assigns a mathematical value for S or O.

Table 2. A scorecard correctly predicts the transitive distribution. S or O is given +/-2 points in the initial position, +/-1 point in the middle position, and zero points in the final position, while the verb (in brackets) is neutral. A new generalization is based on granting decreasingly positive points (from left to right) to the subject and the object in SO languages and the corresponding negative points in OS languages.

Order	Points for position			Total
- SO -	+2	+1	+0	
SOV	S	O	(V)	+3
SVO	S	(V)	O	+2
VSO	(V)	S	O	+1
- OS -	-2	-1	-0	
VOS	(V)	O	S	-1
OVS	O	(V)	S	-2
OSV	O	S	(V)	-3

- (ii) This value systematically decreases from left to right.
- (iii) If SO, the value is positive.
- (iv) If OS, the value is negative.

The number of rules matters when predictive models are evaluated on the principle of parsimony. Manning and Parker (1989) also predict the correct distribution but with a complex algorithm. According to their proposal,

- (i) Constituents are ideally ordered from smallest to largest.
- (ii) S is small.
- (iii) O is medium.
- (iv) V is large.
- (v) The first constituent from the left is assessed first.

Thus, S-initial (small first) is preferred over V-initial (large first),

- (vi) However, O is not independent but bonded with the next constituent.
- (vii) O-bonding makes the O-initial orderings (OV)S and (OS)V front-larger than other types regardless of what is bonded with O (in brackets).
- (viii) The rest of the sentence is assessed second.

Favouring small before large in step (viii), as ordered in their (i), Manning and Parker arrive at the same six-way distribution as Table 2. Their algorithm is broken down into eight rules, four more than the novel one above. Much of the difference depends on the three stipulations defining the constituency sizes (S=small, O=medium, V=large), which are coincidentally the reverse of Hawkins's (1994) claim about SOV heaviness (3.1). If these were proved to be a typological fact, they might be taken for granted, thus reducing the full account to five statements. But Song (1991, 95) specifically

points out that Manning and Parker's stipulations are not supported by typological evidence.

Moreover, there is another possible set of four rules that generates the same pattern as [Table 2](#).

- (i) Each position in the transitive clause assigns a mathematical value for V.
- (ii) This value systematically increases from left to right.
- (iii) If SO, the value is positive (thus, e.g., +3 points for SOV).
- (iv) If OS, the value is negative (thus, e.g., -3 points for OSV).

The reason is that these two formulations are logically equivalent. However, the data suggest that the place of V itself is not significant because V-final is the most and least preferred, V-medial is the second-most and second-least preferred, and V-initial is the third-most and third-least preferred. Thus, the place of V appears to be a by-effect of the orientation of S and O in the clause. Still, there may be potential here for supplanting [Table 2](#). In such a case, the question arises of why V should be the decisive factor based on the research discussed in the following sub-sections, especially (4.4). I have not found any studies that bring forward empirical evidence, whether from typology, language acquisition, or neuroscience, demonstrating how V is the determinant variable of the three. Therefore, my overall model of explanation (4.2) proposes [Table 2](#), which is compatible with research relating to markedness and the OS/SO variable.

4.2 *A theory for the generalization*

What lies beneath [Table 2](#)? A parsimonious explanatory framework proposed here builds on Jakobson (1963), who argued that all OS orderings are emphatic. The implication is that people tend to perceive frontal O as marked or topicalized rather than basic ([Section 3](#)). This notion could be extended to the correct generalization S-initial > V-initial > O-initial. The verb is neutral in this respect, as we see from the full distribution where V-initial, V-medial, and V-final orderings are equally preferred and unpreferred.

The rising dynamism pattern $S < V < O$ is familiar from the communicative dynamism (CD) of the Prague linguistic circle, which explains it on the basis of a variable degree of communicative importance within the information flow, following from the narrative where new information is constantly being added to established information. Thus, the subject contains old information more often and is therefore less marked or less 'dynamic' (Firbas 1971; 1987; Davidse, Van Praet, and Njende 2019).

However, the explanatory model proposed here lays a further emphasis on early communication or HIGHLIGHTING the distinction between the two NPs (S and O). Thus, the SO preference is constructed in terms of fronting S before O (SO > OS) and reaffirmed in its relation to V (SOV > VSO). There is a preference to front the fronting solution of the communication problem between the two nominal phrases as if beginning by making clear: **this is this, that is that** (S versus O), and then communicating their relationship through the verb. SVO is intermediate between the two because the solution becomes obvious from the start, but it does not highlight the this-over-that part and is therefore not equally apposite. Table 2 is important in this respect because it demonstrates that the hierarchy of SOV > SVO > VOS is mathematically correct.

Regarding markedness theory, previous criticism needs to be considered. Haspelmath (2006) argues against markedness as a term due to its ambiguity. Based on Haspelmath's taxonomy, the concept exploited here relates to "MARKEDNESS AS ABNORMALITY", especially Haspelmath's type 10 "Distributional markedness: markedness as restricted distribution". The present paper points out that the object is restricted to the transitive clause, and therefore has a restricted distribution, whereas the subject appears in both intransitive and transitive clauses. Thus, if in a given text of a hundred clauses, one-fifth of the full clauses are transitive, one expects to find 100 subjects and 20 objects, exemplifying the distributional restrictedness of the object.

The actual number of subjects in many languages might be lower, especially in stories and other narratives, due to different manifestations of 'pro-drop' (i.e., omission of the pronoun), but they are still likely to be more frequent. Another crucial factor to consider is that in stories and narratives, when the agent is not repeated, it is IMPLICIT in the clause. This rather supports the idea of a more frequent normalization of the subject (or agent as discussed in Section 2): it may be simply given (i.e., zero-marked). In such terms, too, the object (i.e., patient) is more abnormal and more marked. Such issues are unlikely to be controversial but relate to the question of how UNMARKEDNESS should be extracted from corpora.

Greenberg considered frequency as the main determinant of markedness in grammar. He proposed that unmarked categories could be identified based on "the frequency of association of things in the real world" (Battistella 2015, 534; Greenberg 1966, 66).

The relation of frequency and SALIENCE, i.e., being prominent or standing out from the rest, can be illustrated by imagining visiting a bookstore in New York to inquire in what languages they have Gurnah's latest novel. If the clerk informs that it is available in "French and English", the answer is fine, but it could be interpreted as bringing attention to the fact that it is

available in *French*. If they say it is available in “English and French”, it seems just normal. A cursory Internet search shows “*English and French*” to be the preferred order in English but “*français et anglais*” in French, which shows that such normality is not based on alphabetic order. The more usual item, based on experience, tends to be first in the unmarked order.

The typological data (Figure 2) are compatible with the notion that the object tends to be perceived as emphatic when fronted and is therefore postponed in the dominant order. The effect is intensified when the relationship between O and S is highlighted by fronting. By contrast, OSV, which highlights OS, is the most marked ordering and therefore the most unpreferred as the basic order. As pointed out by Jakobson (1963), there is a focus at the beginning of the sentence. Based on Table 2, such tension gradually eases when the verb moves to the front, shifting to a positive tension—think of it in terms of statistically justified discrimination—as the order is flipped to SO. SOV, which highlights SO the most, is the most predictable ordering.

Alternatively, the phenomenon can be viewed from the perspective of information structure and information status. Subjects tend to contain topical information, which becomes given in the discourse, and objects tend to add new information as comments about the subject. As pointed out in (3.1), this is also what Ancient thinkers meant with PREDICATION (as relates to the subject-predicate divide), and it suffices to explain the tendency for SO > OS, but the concept of fronting such a solution is required for the prediction of the whole distribution, whereby it is best described in terms of highlighting, nonetheless. It is not equally clear why the topic-comment relationship would need to be brought to the front. In Praguian terms, topicalization is thought to favour SVO, which is especially beneficial for European relativization with relative pronouns, which ideally immediately follow their antecedent, suggesting a preference for O-final, especially SVO. It seems more plausible that widely discussed issues such as information structure and animacy further contribute to the secondariness of O but do not change the conclusion based on highlighting.

4.3 Type-based versus gradient approaches

Levshina (2019) and Levshina et al. (2023) argue for abandoning dominant-order typology in favour of a gradient approach. Their gradient corpus analyses are interesting, but relate especially to morphological marking, linking more complexity with free word order and less complexity with more static order. Thus, they might rather relate to what was said in Section 2 about returning to the question of NO-DOM when the main pattern is accounted for.

The proposal of taking intra-linguistic variation into account, according to the authors, derives from “the emergentist usage-based” view. The

underlying idea that word order is not “rigid” is, however, much older as seen from the proponents’ sources as well as this paper. Levshina et al. argue that statistical thresholds or cut-off points defining types are arbitrary. However, as briefly discussed in [Section 2](#), the inclusion criteria for S/V/O typology are mostly clear. Languages can be classified based on the 2:1 ratio. In practice, specialists of a given language tend to be based on a tradition of considering it as falling into one of the types; e.g., Dutch and German SVO/SOV, which is based on the unmarked order in the main clause versus the sub-clause. Much of the overall variation depends on the sentence type, e.g., interrogatives typically involve inversion in Europe, and there are many kinds of special cases among the World’s languages.

Here, the pertinent question is simply: How do speakers of language X normally express $P(x,y)$? The purpose of the question is to find a basic transitive order, and most languages have one regardless of the presence of variation. With such respect, the Dutch and German SVO/SOV may merely relate to pedagogical heuristics, and they should rather be considered SVO in transitive typology. To clarify this, it is not the case in Dutch that the sentence ‘The fox chased the rabbit’ normally translates to *De vos (S) het konijn (O) achtervolgde (V; SOV)*, it is not the basic order of the standalone statement, which should be SVO. Uncertainty in the criteria for a small number of languages may be confusing. These languages, however, also representing NO-DOM, do not count here (cf. [Figure 2](#)).

Let us also consider what Croft wrote about rigidity in a source cited by the authors. In the context of “rigid SOV”, Croft (2003, 78) discusses Greenberg’s universals 6, 7, 13, and 15. He concludes that dominant SOV most frequently alternates with OSV, while dominant VSO alternates with VOS. The observed variation suggests SO to OS inversion in both types, which is likely to be related to the topicalization of O. This brings us back to the starting point of this paper and the issue of secondariness as discussed in (4.2): the OS variant is emphatic and topicalized in both systems. But gradient statistics only show that there is variation. It was through acknowledging dominant SOV and VSO that we realized that the most predictable variant for both types was simple OS inversion (**SOV** → **OSV**, and **VSO** → **VOS**), a research concept in its own right. Such simplicity cannot work out for SVO languages, where the constituents are separated by the verb, rendering the most straightforward solution inapplicable as an inherent property of this structural type. It leads to the further question of what, then, is the basis of variation in SVO. Such issues demonstrate that dominant word order leads to an understanding of intra-linguistic expectations and problematics, and further research hypotheses.

4.4 Empirical evidence

Highlighting ‘this over that’ is yet to be subjected to further study. Currently, though, it can be demonstrated to be compatible with findings from various fields.

A moderate tendency to mention the most frequent item first has long been known in educational psychology (Bowers 1932), and repetition of information typically leads to memory consolidation (Smith 1968). In social sciences, including sociolinguistics, the NORM-DIVERGENT DICHOTOMY is associated with salience. Frequency is a typical factor contributing to salience. At the neuroscientific level, salience relates to the core function of the brain to select the most relevant data from the information flow (Uddin 2015). In cognitive-pragmatic linguistics, it relates to the management of attention (Grabski and Frohning 2008). These questions shed light on how this-over-that highlighting might arise subconsciously from the statistics of human experience including communication.

Boswijk and Coler (2020, 7) argue that salience is an ambiguous term because it can refer to either high or low frequency. The more frequent feature is more familiar and easier to process, but infrequency results in a degree of unexpectedness. Neuroscientists (Zarcone et al. 2016) have isolated the two contrasting principles into PREDICTABILITY versus SURPRISAL despite admitting that there is still much research to be done. On a preliminary basis, these concepts might explain the two modes of agent-patient highlighting. When a predictable item comes first, the order is neutral (‘salience as predictability’), but when an infrequent item comes first, it tends to be perceived as bringing focus to the exception (‘salience as surprisal’).

In some languages, topical inversion occurs without added morphological marking, whereby the subject and the object are inferred from the context. In modern Kaqchikel (Mayan, Guatemala), the translation for ‘The carpenter cut the tree’ can appear in the canonical VOS order, but also in VSO as well as in SVO with a topicalized subject. Sentences where the subject and object are semantically reversible, such as ‘The cat chased the dog’ (or vice versa), are ambiguous between VOS and VSO. However, the VOS interpretation is overwhelmingly preferred. This raised the question of how it relates to processing efficiency theories, which are generally thought to provide empirical evidence for current syntactic theories. Brain imaging techniques were used during the processing of transitive sentences of Kaqchikel, revealing that the speakers were best at processing the canonical VOS order. Koizumi and Kim (2016) conclude that processing preferences appear to be language-specific rather than reflecting a universal preference for SO ordering. The researchers (Koizumi et al. 2014) propose INDIVIDUAL GRAMMAR THEORY, arguing that processing preferences are primarily due to

grammatical factors (including grammatical complexity) in individual languages. Importantly, Koizumi and Kim (2016) conclude that processing efficiency does not explain the statistical prevalence of SO over OS languages.

The problem with the individual grammar explanation is that language-individual reasons may not be universalizable cross-linguistically. It is appropriate to point out that this-over-that highlighting proposed in the present paper is universalizable and based on all of the applicable data. It is simple and compatible when such highlighting is not to be taken as an exceptionless rule but as a tendency, as precisely manifested in typology. Ultimately, it is not claimed that the highlighting theory is proven an indisputable truth, only that it entails a simpler and more accurate explanatory model of the empirical data and therefore supplants other theories as is standard scientific practice.

Finally, in the context of language acquisition research, Goldin-Meadow et al. (2008) conducted a series of clinical studies of nonverbal order suggesting that SOV is the preferred semantic ordering of events regardless of one's native language. The authors also refer to cases where sign languages (Nicaraguan, and Al-Sayyid Bedouin), which were independently developed by the deaf communities, have SOV in contrast to their areal SVO. The researchers suggest that their results may support the Proto-World SOV hypothesis because people conceptualize events in that order pre-linguistically. Goldin-Meadow et al. do not develop a full six-way hierarchy of preferred and unpreferred orderings, but their results are likewise compatible with the this-over-that highlighting explanation. This theory has sufficient explanatory power to cater to observations made by Goldin-Meadow et al. (2008) and Givón (2018), which suggest a pre-linguistic cognitive bias underpinning human languages. It explains the bias by transferring the statistics of experience of reality to the statistics of communication.

5. Conclusion

This paper introduced a new explanatory model for the crosslinguistic distribution of the transitive clause. The simplest explanation of the cross-linguistic variation of the transitive sentence is based on the notion that there is a tendency to perceive objects as emphatic if they appear (i) sentence-initially (O-initial) or (ii) before the subject (OS). Both cases are mostly avoided as a dominant transitive order among the World's languages. Conversely, if the subject precedes the object (SO), their early position is preferred because it highlights the primacy of the subject over the object. The variable place of the verb in the distribution appears to be a side effect of different grades of highlighting the relationship between the subject and the object. These considerations give rise to the hierarchy

SOV > SVO > VSO > VOS > OVS > OSV, as indicated by a scorecard (Table 2).

At the level of the distribution pattern, the attested frequency of SOV over SVO languages is significant, albeit not dramatic. Their relative similarity is most likely a historical accident. It has been demonstrated that SOV is clearly the most common ordering across language families. The cognitive and universalist explanation proposed in this paper exploits markedness theory as proposed by Roman Jakobson at the 1961 seminal Dobbs Ferry conference on linguistic universals, further drawing from Nikolai Trubetzkoy's (1890–1938) notion of frequency as an explanatory factor of markedness, which was advocated by Joseph Greenberg and Charles E. Osgood.

The present paper exploits the concept of the statistically marked object as the second in order, giving a unique explanation of the whole distribution pattern. The object is second to the subject in terms of frequency and, as a solution to the linearization problem, it is marked as being so in the basic word order. Such markedness is, again, marked or HIGHLIGHTED by early SO and diminished by late OS. These bring about a preference for early SO and late OS.

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